Ahmed Ezzat Ahmed

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Professional Summary

AI Engineer specializing in Arabic Speech & Voice AI. TTS Lead at hams.ai, building and operating a production real-time Saudi Arabic TTS system powering live enterprise voice agents – owning the stack end to end, from 13.6k hours of training data and F5-TTS fine-tuning to TensorRT-optimized streaming inference at 154 ms time-to-first-audio. 2+ years shipping dialect-specific TTS/ASR systems (Saudi, Egyptian, MSA) on production GPU infrastructure.

Experience

AI Engineer (TTS Lead, Voice AI Team), hams.ai — February 2026 – Present

- Built and operate "Hams Turbo TTS," a production real-time Saudi Arabic text-to-speech system powering live enterprise voice agents in the Gulf market; own the stack end to end, from data and training through deployment and reliability.
- Fine-tuned F5-TTS for Saudi Arabic across multiple model generations on ∼13,600 hours of speech, training on multi-GPU cloud infrastructure (A100/H100).
- Delivered a ∼9x combined inference speedup: ∼4.5x via pruned-step sampling , plus ∼2x from a custom TensorRT FP16 backend on production L4 GPUs.
- Achieved ∼154 ms mean / 253 ms p95 time-to-first-audio in production (RTF 0.03–0.05); benchmarked 50 concurrent requests with zero errors on a single L4 GPU.
- Own the end-to-end Cloud production deployment (Docker, TensorRT, L4) with zero-downtime cutovers: automated pre-cutover quality gates, instant auto-rollback, and graceful readiness-gated shutdown, cutting cold-start from ∼43 s to 7.5 s.
- Designed a server-side Arabic text-normalization layer (Saudi number words, acronym spelling, spoken emails/URLs, brand pronunciations).

AI Engineer, GenArabia — September 2025 – January 2026

- Fine-tuned the open-source Spark-TTS model for Arabic on a custom-curated 150-hour dataset using H100 GPUs, engineering it to handle diacritics and publishing the result to the Hugging Face community.
- Fine-tuned Whisper Large V3 and NVIDIA Parakeet-TDT models for Air Traffic Control (ATC) communications, achieving 60% reduction in Word Error Rate (WER) through domain-specific training, data augmentation, and a 6-gram KenLM language model.
- Designed a synthetic speech data generation pipeline for telecom use cases, including telephony codec simulation (G.711/AMR-NB) and real-to-synthetic data mixing strategies.
- Engineered end-to-end speech processing pipelines for high-fidelity TTS and STT systems, with serverless deployment on Modal behind FastAPI endpoints.

AI Engineer, Andalusi — June 2025 – August 2025

- Delivered a quantized background-removal model (75% size reduction) for edge deployment on mobile, and shipped a serverless object-removal endpoint that drove premium-tier subscription growth.

AI Engineer, E Connect Africa — June 2024 – May 2025

- Owned the full lifecycle of TTS for a course-generation product: sourced and prepared a 25-hour Arabic speech dataset, trained Tortoise, XTTS v2, and Fish Speech models, and deployed them with high vocal naturalness.
- Built end-to-end audio-visual sync pipelines integrating lip-sync models (SadTalker, Wav2Lip, LatentSync) with TTS outputs for realistic speech-driven character animation, optimized for minimal hardware usage in interactive learning content.

Machine Learning Intern, TensorGraph — July 2023 – August 2023

- Built a BERT-based entity-recognition system for restaurant phone-order automation; used GPT for synthetic data generation, cutting annotation costs by 80%.

Publications

Lightweight Plant Leaf Classification Based on Transfer Learning: A Comparative Study 2024

- On-device plant species identification with lightweight transfer-learning models. DOI: 10.1109/JAC-ECC64419.2024.11061212

Education

Helwan University, BS in Computer Science 2024

- Cumulative GPA: 3.27 (Very Good). Graduation project (EduAI): YOLOv8-based prohibited-object detection (86 mAP) plus eye-gaze tracking and a Siamese face-recognition network for online-exam proctoring – reduced cheating incidents by 89%.

Selected Projects

Large-Scale Saudi Arabic Speech Corpus (Hugging Face) 2026

- Designed and executed the full data pipeline for a 1,368-hour Saudi/Khaleeji Arabic corpus targeting TTS and ASR: source curation, multi-engine transcription with CER-based quality control, diacritization, and dialect annotation.

Real-Time Audio Chat Application with Voice AI Pipeline 2025

- Production-ready voice agent integrating VAD, Whisper Large v3 Turbo (STT), Qwen 3 (LLM), and Resemble AI Chatterbox (TTS) over WebRTC, with low-latency streaming and barge-in for natural mid-response interruption.

EgyLens – VQA for Egyptian Arabic (Kaggle) 2025

- Fine-tuned Google's Gemma 3n on Egyptian Arabic VQA datasets for on-device, privacy-preserving visual understanding with offline inference.

Volunteering

Manus Fellow, Manus AI July 2025 – Present

- One of 20+ global Fellows representing Manus in Cairo; organized 4 community events reaching 200+ participants – most recently a "Build Your Startup in 4 Hours" hackathon (July 2026) – and grew a community of 3,000+ active members. Currently Chairman of MIE 2026 (IEEE Young Professionals Egypt), coordinating mentor recruitment across 345 engineering project teams.

Skills

Speech & Voice AI: Arabic dialect TTS/STT (Saudi, Egyptian, MSA), F5-TTS, Kokoro, Chatterbox, XTTSv2, Spark-TTS, RVC, Whisper, NVIDIA Parakeet-TDT, NeMo, Silero VAD, pyannote, forced alignment (MFA, CTC), librosa, torchaudio, KenLM

Voice Agents & Real-Time Pipelines: pipecat (incl. custom TTS plugin development), LiveKit, WebRTC, WebSocket streaming, FastAPI, VAD/STT/LLM/TTS orchestration, barge-in, end-of-utterance detection

Machine Learning: PyTorch, TensorFlow, fine-tuning, flow-matching models, Diffusion Models, model distillation, State Space Models, Transformer architectures, LoRA/PEFT, model quantization

Deployment & Infrastructure: GCP (NVIDIA L4 / A100 / H100 / B200), TensorRT, Docker, Modal, RunPod, Nginx, Hugging Face Hub, zero-downtime deployment pipelines, on-prem GPU deployment, license enforcement & usage metering for enterprise delivery

Languages & Tools: Python (primary), JavaScript/TypeScript, Bash, Git/GitHub, WSL

Soft Skills: Teamwork, time management, research analysis, technical writing, ownership in startup environments